

# E-COMMERCE RETURN RATE REDUCTION ANALYSIS

## *Project Report — Diagnostic Analysis, Return-Risk Scoring & Interactive Dashboard*

Report date: July 29, 2026 | Data scope: 250,000 customer transactions.

### 1. Executive Summary

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This project investigates why customers return products in an e-commerce environment and how return behavior varies across product categories, countries, payment methods, and marketing channels. The analysis pipeline spans four components: a relational SQL schema that models orders, products, and returns; a Python/NumPy exploratory analysis of a 250,000-row customer transaction dataset; a machine-learning-derived Return-Risk score used to flag high-risk products; and a two-page interactive Power BI dashboard with drill-through filters for country, category, and marketing channel.

The headline finding is that the primary transactional dataset (`ecommerce_customer_data_custom_ratios.xlsx`) is return-neutral: the overall return rate is 49.8% among orders with a recorded return status, and this rate is essentially flat across product category (49.6%–49.9%), gender, and payment method. In other words, in this dataset return behavior does not vary meaningfully by the segment dimensions typically assumed to matter. This is an important and actionable data-quality finding in itself, because it means category or channel-based return-reduction strategies built on this file alone would not be well supported by evidence.

Where real, differentiated signal does exist is in (a) the smaller demonstration schema (`Ecommerce.sql` / `Online_Sales_Results.xlsx`), which shows Electronics returned at a much higher rate than Fashion or Furniture and Instagram-sourced orders returning at 100%, and (b) the exported `high_risk_products.csv`, which isolates 75,137 SKUs with a model-assigned Return-Risk score of 0.71 or higher (mean 0.82) — a working shortlist operations teams can act on directly, even though the score itself does not vary strongly by category or price in the current model. Section 5 and Section 7 unpack both findings and what they imply for next steps.

### 2. Business Objective & Scope

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E-commerce returns erode margin through reverse logistics, restocking, and lost sale value. The business objective of this project is to (1) quantify the overall and segment-level return rate, (2) identify the stated reasons customers return items, (3) build a repeatable way to flag high-risk products before or shortly after purchase, and (4) give merchandising and operations stakeholders a self-serve dashboard to explore returns by country, category, and channel without needing a new SQL query each time.

#### In scope

- Descriptive analysis of return rate by product category, payment method, gender, and churn status
- A relational schema (Orders, Products, Returns) with return-reason level SQL queries
- A Return-Risk scoring export used to generate a high-risk product watchlist
- A two-page Power BI dashboard with KPI cards, trend charts, and drill-through slicers

#### Out of scope

- Root-cause survey or qualitative interviews with customers who returned items
- Financial quantification of return cost (shipping, restocking, write-off) — not present in the source data

- Real-time / streaming scoring; the current pipeline is a batch, point-in-time export

### 3. Data Sources & Project Architecture

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The project is built from six deliverables, each covering a different stage of the analysis pipeline:

| File                                       | Role in pipeline                                                                                 | Size                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|
| Ecommerce.sql                              | Schema design (Orders / Products / Returns) + demo return-reason & category return-rate queries  | 8 orders, 5 products, 4 returns |
| Ecommerce.ipynb                            | Python/NumPy/pandas EDA notebook: overall & category return-rate calculation, bar and pie charts | 14 cells                        |
| ecommerce_customer_data_custom_ratios.xlsx | Primary transactional dataset used for the EDA                                                   | 250,000 rows × 13 columns       |
| high_risk_products.csv                     | Model output: products with a Return-Risk score, filtered to the high-risk band                  | 75,137 rows × 4 columns         |
| Online_Sales_Results.xlsx                  | Exported result set from the SQL demo schema (order/return/category detail)                      | 8 rows × 7 columns              |
| E-Commerce_Analysis_Dashboard.pbix         | Power BI dashboard: KPIs, trend and category visuals, drill-through slicers                      | 2 report pages                  |

*Table 1. Source files and their function in the analysis pipeline.*

#### Primary dataset schema

The main transactional file contains: Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Customer Age, Returns (0/1/blank), Customer Name, Age, Gender, and Churn. Product Category takes four values — Books, Clothing, Electronics, Home.

### 4. Methodology

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#### 4.1 SQL layer

A normalized schema (Orders → Products, Orders → Returns) supports joins that compute return rate by category and by country, and a listing of order-level return reasons. This layer was validated on a small, hand-built demo dataset (8 orders) to confirm the join logic and rate-calculation formulas before being applied conceptually to the larger file.

#### 4.2 Python exploratory analysis

The notebook loads the 250,000-row dataset with pandas, converts the Returns column to a NumPy array, and computes an overall and per-category return rate. During review of this project, a data-typing issue was identified: the notebook's original logic compared the Returns values against the strings 'yes', 'returned', 'true', and '1', but the column is stored as a float (0.0 / 1.0 / blank), so str(1.0) evaluates to "1.0" and never matched — this produced a false reading of 0.00% return rate in the saved notebook output. For this report, the return rate was recomputed directly from the numeric column (treating 1.0 as returned, 0.0 as not returned, and blanks as missing) to give an accurate

picture; see Section 5 for the corrected figures. This is flagged explicitly because it is a realistic, common failure mode — string-vs-numeric flag comparisons — worth guarding against with an assertion or dtype check in production code.

### 4.3 Return-risk scoring

high\_risk\_products.csv is the output of a scoring step (not included as a notebook cell, but present as an artifact) that assigns each product a Return\_Risk value between 0 and 1 and exports rows above an approximate 0.70 threshold. This functions as a binary classifier's high-confidence output band: 75,137 of the scored SKUs cleared the threshold, and of those, 74,658 (99.4%) were in fact returned — indicating the score is a strong filter for isolating already-returned or return-prone inventory, even though (as shown in Section 6) it does not yet differentiate risk within category or price bands.

### 4.4 Dashboard

The Power BI file contains two report pages. Page 1 combines KPI cards, a KPI visual, two slicers, an area chart, a clustered bar chart, a donut chart, a scatter chart, a funnel, and an action button — built for at-a-glance monitoring with drill-through by category/country/channel. Page 2 is trend-and-composition focused: an area chart, a waterfall chart, a combo line/stacked-column chart, a treemap, a slicer, and a line chart — built for period-over-period and category-mix analysis.

## 5. Key Findings

### 5.1 Overall return rate (primary dataset, corrected)

| Metric                                          | Value          |
|-------------------------------------------------|----------------|
| Total transaction rows                          | 250,000        |
| Rows with a recorded return status              | 202,404        |
| Rows with missing return status                 | 47,596 (19.0%) |
| Return rate — of records with a status          | 49.79%         |
| Return rate — missing treated as "not returned" | 40.31%         |

Table 2. Overall return rate under two reasonable treatments of missing values.

Nearly half of all orders with a known outcome were returned. This is materially higher than typical industry return rates (commonly cited in the 8–15% range for general e-commerce, higher for apparel), which is worth flagging to stakeholders as either a genuine, severe problem or a signal that this synthetic/sample dataset over-represents returns for demonstration purposes.

### 5.2 Return rate by product category

| Category    | Return rate |
|-------------|-------------|
| Books       | 49.89%      |
| Electronics | 49.86%      |
| Home        | 49.84%      |
| Clothing    | 49.60%      |

Table 3. Category is essentially not a differentiator in the primary dataset — a 0.29-point spread across all four categories.

5.3 Return rate by payment method and gender

Payment method shows a similarly narrow spread: Cash 50.12%, PayPal 49.81%, Credit Card 49.79%, Crypto 49.03%. Gender shows Male 49.86% vs. Female 49.72%. The correlation between Returns and Churn is 0.002 — effectively zero, meaning a returned order is not a meaningful predictor of churn in this dataset, and average purchase amount and customer age are also statistically indistinguishable between returned and non-returned orders (differences under 1%).

5.4 Return reasons and category effects (demo schema)

The SQL demo dataset — a small, hand-built set of 8 orders used to validate the schema and query logic — does show clear differentiation, and illustrates what the queries will surface once run against a larger, real order-return dataset:

| Category    | Orders | Returns | Return rate |
|-------------|--------|---------|-------------|
| Electronics | 4      | 3       | 75.0%       |
| Fashion     | 3      | 1       | 33.3%       |
| Furniture   | 1      | 0       | 0.0%        |

Table 4. Category return rate in the demo schema (n=8 orders; illustrative of query logic, not statistically significant at this size).

Return reasons in the demo set: Damaged (2 of 4 returns, 50%), Wrong Size (1, 25%), Changed Mind (1, 25%). By country: USA 75%, Canada 50%, UK 0%. By marketing channel: Instagram 100%, Facebook 50%, Email 50%, Google 0%. These patterns — Electronics skewing toward "Damaged" and Fashion toward "Wrong Size" — are directionally consistent with common e-commerce return drivers (fragile/high-value goods damaged in transit; apparel sized incorrectly) and are exactly the relationships the SQL layer and dashboard are designed to surface at full scale.

6. High-Risk Product Segmentation

high\_risk\_products.csv isolates products scored at or above a ~0.71 Return-Risk threshold — a practical watchlist for merchandising or QA teams.

| Metric                                      | Value                   |
|---------------------------------------------|-------------------------|
| Products flagged as high-risk               | 75,137                  |
| Mean Return-Risk score                      | 0.82 (range 0.71–0.99)  |
| Share of flagged products actually returned | 99.4% (74,658 / 75,137) |
| Category mix — Books                        | 22,509 (30.0%)          |
| Category mix — Clothing                     | 22,415 (29.8%)          |
| Category mix — Electronics                  | 15,169 (20.2%)          |
| Category mix — Home                         | 15,044 (20.0%)          |
| Correlation: Return-Risk vs. Product Price  | -0.012 (negligible)     |

Table 5. High-risk product export summary.

Two things stand out. First, the flag is precise — 99.4% of the products above threshold were, in fact, returned — so it is safe to route this list directly into an operational workflow (e.g., pre-shipment QA hold, targeted packaging upgrades, or a returns-prevention email at checkout). Second, the risk score itself is nearly flat across category (0.8165–0.8176) and uncorrelated with price, which means the underlying model is not yet capturing category- or price-driven risk factors — the score is closer to a binary returned/not-returned flag than a true risk gradient. Strengthening the feature set (e.g., adding return reason, shipping method, review sentiment, or size/fit indicators) would let the score better prioritize within the 75,137-item list rather than treating most of it as uniformly "high risk."

## 7. Interactive Dashboard Overview

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E-Commerce\_Analysis\_Dashboard.pbix delivers the findings above as a self-serve, two-page Power BI report so stakeholders can filter without writing SQL.

### Page 1 — Returns Overview

- KPI cards and a KPI visual for headline metrics (orders, returns, overall return rate)
- Slicers for drill-through filtering (category, country, and/or channel)
- Area chart for return trend over time
- Clustered bar chart for return rate by category
- Donut chart for return-reason mix
- Scatter chart for price vs. return-risk relationship
- Funnel visual for order → return conversion

### Page 2 — Trend & Composition

- Area and combo (line + stacked column) charts for period-over-period return volume
- Waterfall chart to decompose changes in return rate between periods or segments
- Treemap for category/product return-volume composition
- Line chart and slicer for time-based drill-down

Because the dashboard is built on the same field structure documented in Section 3, refreshing it against a return dataset with genuine category/channel variance (rather than the return-neutral 250K file) will make the drill-through filters immediately more informative — the visuals and interactions are already in place.

## 8. Data Quality Notes & Limitations

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- The 250,000-row primary dataset shows almost no variation in return rate across category, gender, or payment method (all within ~1 percentage point of 49.8%), which is atypical for real-world return data and should be validated against the source system before being used for segment-level business decisions.
- 19.0% of return-status values are missing; the two treatments in Table 2 (49.79% vs. 40.31%) bound the true rate but the report does not know why status is missing — e.g., in-transit orders vs. a data-entry gap — and that distinction changes the interpretation.
- The demo SQL/Online\_Sales\_Results dataset (n=8 orders) is useful for validating query logic and is directionally realistic, but is far too small to be statistically reliable on its own.

- The Return-Risk score in `high_risk_products.csv` is highly accurate at separating returned from non-returned (99.4%) but does not yet vary meaningfully by category or price, limiting its use for prioritization within the high-risk list.
- No cost data (shipping, restocking, refund processing) is present in any source file, so return rate cannot yet be translated into a dollar-impact estimate.

## 9. Recommendations

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- Validate the primary dataset against the live order/return system to confirm whether the ~50% return rate and flat category distribution are real or an artifact of how this extract/sample was generated.
- Fix the return-flag comparison logic in the notebook (compare the numeric column directly, or cast with an explicit dtype check) so future refreshes don't silently report 0% due to a string/float mismatch.
- Enrich the risk model with features that are known return drivers — return reason history, shipping carrier/damage rate, size-and-fit signals for apparel, and review sentiment — to convert the current near-binary Return-Risk score into a genuine risk ranking.
- Prioritize the 75,137-item high-risk list for a pilot intervention (e.g., reinforced packaging for Electronics, a size-guide prompt for Clothing at checkout) and measure the before/after return rate for just that pilot segment.
- Add a cost field (unit shipping + restocking cost) to the schema so the dashboard can report return rate in dollar terms, not just order counts, which will make the business case for any intervention much easier to prioritize.
- Once the underlying data issues above are resolved, re-point the Power BI dashboard at the corrected extract — no rebuild of the visuals themselves should be required.

## 10. Conclusion

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This project delivers a complete, working return-analysis pipeline — schema, EDA, a risk-scoring export, and a two-page interactive dashboard — and the underlying mechanics (joins, rate calculations, drill-through filters, high-risk flagging) are sound and demonstrated correctly on the validation data. The main finding from this review is that the large primary dataset does not currently show the kind of category- or channel-level variation that would normally drive a returns-reduction strategy, while the smaller demo schema and the high-risk product export both show the sharper, more actionable patterns (Electronics/damage, Fashion/sizing, a highly precise but not yet well-differentiated risk score) that the next iteration of this project should be built around. Addressing the data-quality items in Section 8 is the highest-leverage next step before acting on category-specific recommendations.